AI-Powered Smart News Intelligence & Fake-News Risk Analyzer

```
!pip -q install -U transformers torch sentencepiece gradio pandas numpy
matplotlib seaborn scikit-learn
```

```
import re
import warnings
import torch
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from collections import Counter

from transformers import (
    pipeline,
    AutoTokenizer,
    AutoModelForSeq2SeqLM
)

warnings.filterwarnings("ignore")

print("PyTorch version:", torch.__version__)
print("CUDA available:", torch.cuda.is_available())

if torch.cuda.is_available():
    print("GPU:", torch.cuda.get_device_name(0))
else:
    print("Running on CPU")
```

```
PyTorch version: 2.11.0+cpu
CUDA available: False
Running on CPU
```

```
print("Loading sentiment analysis model...")

sentiment_model = pipeline(
    "sentiment-analysis",
    model="distilbert-base-uncased-finetuned-sst-2-english"
)

print("Sentiment model loaded successfully!")
```

```
Loading sentiment analysis model...
Loading weights: 100%                104/104 [00:00<00:00,   7.91it/s]
Sentiment model loaded successfully!
```

```
print("Loading news classification model...")

classifier_model = pipeline(
    "zero-shot-classification",
    model="facebook/bart-large-mnli"
)

print("News classification model loaded successfully!")
```

```
Loading news classification model...
Loading weights: 100%                515/515 [00:00<00:00, 991.13it/s]
News classification model loaded successfully!
```

```
print("Loading summarization model...")

summarizer_model_name = "sshleifer/distilbart-cnn-12-6"

summarizer_tokenizer = AutoTokenizer.from_pretrained(
    summarizer_model_name
)

summarizer_model = AutoModelForSeq2SeqLM.from_pretrained(
    summarizer_model_name
```

```python
)

device = torch.device(
    "cuda" if torch.cuda.is_available() else "cpu"
)

summarizer_model = summarizer_model.to(device)

print("Summarization model loaded successfully!")
print("Running summarization on:", device)
```

```
Loading summarization model...
tokenizer_config.json: 100%  26.0/26.0 [00:00<00:00, 165B/s]
vocab.json: 100%  899k/899k [00:00<00:00, 3.11MB/s]
merges.txt: 100%  456k/456k [00:00<00:00, 2.11MB/s]
pytorch_model.bin: reconstructing file: 100%  1.22GB / 1.22GB, 80.6MB/s
pytorch_model.bin: downloading bytes:  933MB, 55.5MB/s
model.safetensors: reconstructing file: 100%  1.22GB / 1.22GB, 55.1MB/s
model.safetensors: downloading bytes:  932MB, 56.6MB/s
[transformers] Please make sure the generation config includes `forced_bos_token_id=0`.
Loading weights: 100%  358/358 [00:00<00:00, 1374.29it/s]
Summarization model loaded successfully!
Running summarization on: cpu
```

```python
print("=" * 60)
print("MODEL STATUS")
print("=" * 60)

print("Sentiment Model      : Loaded")
print("Classification Model : Loaded")
print("Summarization Model  : Loaded")
print("Device               :", device)

print("=" * 60)
```

```
============================================================
MODEL STATUS
============================================================
Sentiment Model      : Loaded
Classification Model : Loaded
Summarization Model  : Loaded
Device               : cpu
============================================================
```

```python
def clean_text(text):
    """
    Cleans the input text.
    """

    if not isinstance(text, str):
        return ""


    text = re.sub(
        r"http\S+|www\S+|https\S+",
        "",
        text
    )


    text = re.sub(
        r"\s+",
        " ",
        text
    )


    text = text.strip()

    return text
```

```
sample_text = """
Artificial intelligence is transforming businesses.
Visit https://example.com for more information.
"""

print("Original:")
print(sample_text)

print("\nCleaned:")
print(clean_text(sample_text))
```

```
Original:

Artificial intelligence is transforming businesses.
Visit https://example.com for more information.


Cleaned:
Artificial intelligence is transforming businesses. Visit for more information.
```

```
categories = [
    "technology",
    "business",
    "sports",
    "politics",
    "science",
    "entertainment",
    "health"
]

print("News Categories:")
for category in categories:
    print("-", category)
```

```
News Categories:
- technology
- business
- sports
- politics
- science
- entertainment
- health
```

```
def classify_news(text):

    text = clean_text(text)

    if len(text.strip()) < 5:
        return {
            "category": "Unknown",
            "confidence": 0.0,
            "all_labels": [],
            "all_scores": []
        }

    result = classifier_model(
        text,
        candidate_labels=categories
    )

    return {
        "category": result["labels"][0],
        "confidence": result["scores"][0],
        "all_labels": result["labels"],
        "all_scores": result["scores"]
    }
```

```
test_news = """
Apple has introduced a new artificial intelligence system
designed to improve productivity and user experience.
"""

result = classify_news(test_news)

print("Predicted Category:")
print(result["category"])

```

```
print("\nConfidence:")
print(round(result["confidence"] * 100, 2), "%")
```

```
Predicted Category:
technology

Confidence:
93.95 %
```

```
def analyze_sentiment(text):

    text = clean_text(text)

    if len(text.strip()) < 5:
        return {
            "sentiment": "UNKNOWN",
            "confidence": 0.0
        }

    result = sentiment_model(text)[0]

    return {
        "sentiment": result["label"],
        "confidence": result["score"]
    }
```

```
test_text = """
The new technology has significantly improved
the customer experience.
"""

result = analyze_sentiment(test_text)

print("Sentiment:")
print(result["sentiment"])

print("\nConfidence:")
print(round(result["confidence"] * 100, 2), "%")
```

```
Sentiment:
POSITIVE

Confidence:
99.98 %
```

```
stop_words = {
    "the",
    "is",
    "a",
    "an",
    "and",
    "or",
    "of",
    "to",
    "in",
    "on",
    "for",
    "with",
    "this",
    "that",
    "are",
    "was",
    "were",
    "has",
    "have",
    "from",
    "by",
    "as",
    "at",
    "it",
    "be",
    "will",
    "about",
    "their",
    "its",
    "they",
```

```python
    "these",
    "those",
    "into",
    "than"
}


def extract_keywords(text, top_n=10):

    text = clean_text(text).lower()

    words = re.findall(
        r"\b[a-zA-Z]{3,}\b",
        text
    )

    words = [
        word
        for word in words
        if word not in stop_words
    ]

    frequency = Counter(words)

```

```python
article = """
Artificial intelligence is transforming businesses around the world.
Companies are investing in artificial intelligence, machine learning,
automation, cloud computing and cybersecurity technologies.
"""

keywords = extract_keywords(
    article,
    top_n=10
)

print("Important Keywords:\n")

for word, count in keywords:
    print(f"{word} : {count}")
```

```
Important Keywords:

artificial : 2
intelligence : 2
transforming : 1
businesses : 1
around : 1
world : 1
companies : 1
investing : 1
machine : 1
learning : 1
```

```python
def summarize_text(text):

    text = clean_text(text)


    if len(text.split()) < 40:
        return text


    inputs = summarizer_tokenizer(
        text,
        return_tensors="pt",
        max_length=1024,
        truncation=True
    )


    inputs = {
        key: value.to(device)
        for key, value in inputs.items()
    }


```

```
    with torch.no_grad():

        summary_ids = summarizer_model.generate(
            input_ids=inputs["input_ids"],
            attention_mask=inputs["attention_mask"],
            max_new_tokens=100,
            min_new_tokens=20,
            num_beams=4,
            do_sample=False,
            early_stopping=True
        )


    summary = summarizer_tokenizer.decode(
        summary_ids[0],
        skip_special_tokens=True
    )

    return summary
```

```
article = """
Artificial intelligence is rapidly changing modern businesses.
Companies are adopting machine learning, automation and cloud-based
technologies to improve productivity and customer experiences.
AI systems are also being used in healthcare, finance, education,
manufacturing and cybersecurity. Organizations are investing heavily
in AI research because intelligent systems can process large amounts
of information and support faster decision making.
Researchers are also working to make artificial intelligence safer,
more reliable and more useful for society.
"""

summary = summarize_text(article)

print("=" * 60)
print("ORIGINAL ARTICLE")
print("=" * 60)

print(article)

print("\n" + "=" * 60)
print("AI GENERATED SUMMARY")
print("=" * 60)

print(summary)
```

```
[transformers] Both `max_new_tokens` (=100) and `max_length`(=142) seem to have been set. `max_new_tokens` will take precedence.
[transformers] Both `min_new_tokens` (=20) and `min_length`(=56) seem to have been set. `min_new_tokens` will take precedence. P
============================================================
ORIGINAL ARTICLE
============================================================

Artificial intelligence is rapidly changing modern businesses.
Companies are adopting machine learning, automation and cloud-based
technologies to improve productivity and customer experiences.
AI systems are also being used in healthcare, finance, education,
manufacturing and cybersecurity. Organizations are investing heavily
in AI research because intelligent systems can process large amounts
of information and support faster decision making.
Researchers are also working to make artificial intelligence safer,
more reliable and more useful for society.


============================================================
AI GENERATED SUMMARY
============================================================
 Companies are adopting machine learning, automation and cloud-based technologies . AI systems are also being used in healthcare
```

```
risk_words = [
    "shocking",
    "secret",
    "miracle",
    "guaranteed",
    "breaking",
    "unbelievable",
    "exposed",
    "you won't believe",
```

```
    "100%",
    "must share",
    "urgent",
    "hidden truth",
    "incredible discovery",
    "they don't want you to know"
]


def fake_news_risk(text):

    text_lower = clean_text(text).lower()

    detected_words = []

    for word in risk_words:

        if word in text_lower:
            detected_words.append(word)

    word_count = max(
        len(text_lower.split()),
        1
    )

    # Calculate linguistic signal score
    risk_score = (
        len(detected_words) /
        word_count
    ) * 100

    risk_score = min(
        risk_score * 10,
        100
    )

    if risk_score >= 50:
        level = "HIGH"

    elif risk_score >= 20:
        level = "MEDIUM"

    else:
        level = "LOW"

    return {
        "risk_level": level,
        "risk_score": round(risk_score, 2),
        "signals": detected_words
    }
```

```
test_text = """
BREAKING! This shocking secret discovery is guaranteed
to change everything. You won't believe the hidden truth!
"""

risk = fake_news_risk(test_text)

print("Risk Level:")
print(risk["risk_level"])

print("\nRisk Score:")
print(str(risk["risk_score"]) + "%")

print("\nDetected Signals:")
print(risk["signals"])
```

```
Risk Level:
HIGH

Risk Score:
100%

Detected Signals:
['shocking', 'secret', 'guaranteed', 'breaking', "you won't believe", 'hidden truth']
```

```
def analyze_news(text):

    text = clean_text(text)

    if len(text.strip()) < 10:

        return {
            "error": "Please enter at least 10 characters."
        }


    category_result = classify_news(text)


    sentiment_result = analyze_sentiment(text)


    keywords_result = extract_keywords(
        text,
        top_n=10
    )



    risk_result = fake_news_risk(text)



    summary_result = summarize_text(text)

    return {

        "category":
            category_result["category"],

        "category_confidence":
            round(
                category_result["confidence"] * 100,
                2
            ),

        "sentiment":
            sentiment_result["sentiment"],

        "sentiment_confidence":
            round(
                sentiment_result["confidence"] * 100,
                2
            ),

        "keywords":
            [
                word
                for word, count in keywords_result
            ],

        "risk_level":
            risk_result["risk_level"],

        "risk_score":
            risk_result["risk_score"],

        "risk_signals":
            risk_result["signals"],

        "summary":
            summary_result
    }
```

```
news = """
Artificial intelligence researchers have developed new technologies
that could help hospitals analyze medical data more efficiently.
Scientists said these systems could improve diagnosis and reduce
the time required to process patient information.

Organizations are increasingly adopting artificial intelligence,
```

```
machine learning and cloud technologies to improve productivity.
Researchers are also working on safer and more reliable AI systems.
"""

result = analyze_news(news)

if "error" in result:

    print(result["error"])

else:

    print("=" * 70)
    print("        AI NEWS INTELLIGENCE REPORT")
    print("=" * 70)

    print("\n NEWS CATEGORY")
    print("-------------------------")
    print(result["category"].upper())

    print(
        "Confidence:",
        result["category_confidence"],
        "%"
    )

    print("\n SENTIMENT")
    print("-------------------------")
    print(result["sentiment"])

    print(
        "Confidence:",
        result["sentiment_confidence"],
        "%"
    )

    print("\n KEYWORDS")
    print("-------------------------")
    print(", ".join(result["keywords"]))

    print("\n VERIFICATION RISK INDICATOR")
    print("-------------------------")
    print(result["risk_level"])

    print(
        "Risk Score:",
        result["risk_score"],
        "%"
    )

    print("\nDetected Signals:")
    print(result["risk_signals"])

    print("\n AI GENERATED SUMMARY")
    print("-------------------------")
    print(result["summary"])

```

```
[transformers] Both `max_new_tokens` (=100) and `max_length`(=142) seem to have been set. `max_new_tokens` will take precedence.
[transformers] Both `min_new_tokens` (=20) and `min_length`(=56) seem to have been set. `min_new_tokens` will take precedence. P
======================================================================
        AI NEWS INTELLIGENCE REPORT
======================================================================

 NEWS CATEGORY
-------------------------
TECHNOLOGY
Confidence: 63.2 %

 SENTIMENT
-------------------------
POSITIVE
Confidence: 76.33 %

 KEYWORDS
-------------------------
artificial, intelligence, researchers, technologies, could, more, systems, improve, developed, new

 VERIFICATION RISK INDICATOR
```

```
-------------------------
LOW
Risk Score: 0.0 %

Detected Signals:
[]

 AI GENERATED SUMMARY
-------------------------
 Organizations are increasingly adopting artificial intelligence, machine learning and cloud technologies to improve productivit

======================================================================
```

```
sample_news = [

    {
        "text":
        "The company launched a new artificial intelligence platform for software developers.",
        "category":
        "technology"
    },

    {
        "text":
        "The football team won the championship after an exciting final match.",
        "category":
        "sports"
    },

    {
        "text":
        "The company reported strong financial growth during the quarter.",
        "category":
        "business"
    },

    {
        "text":
        "Scientists discovered a new method for improving renewable energy.",
        "category":
        "science"
    },

    {
        "text":
        "The government announced a new policy for digital infrastructure.",
        "category":
        "politics"
    },

    {
        "text":
        "Doctors developed a new approach to improve patient treatment.",
        "category":
        "health"
    },

    {
        "text":
        "The movie received positive reviews from audiences around the world.",
        "category":
        "entertainment"
    }

]

df = pd.DataFrame(sample_news)

print("Sample Dataset:")
display(df)
```

Sample Dataset:

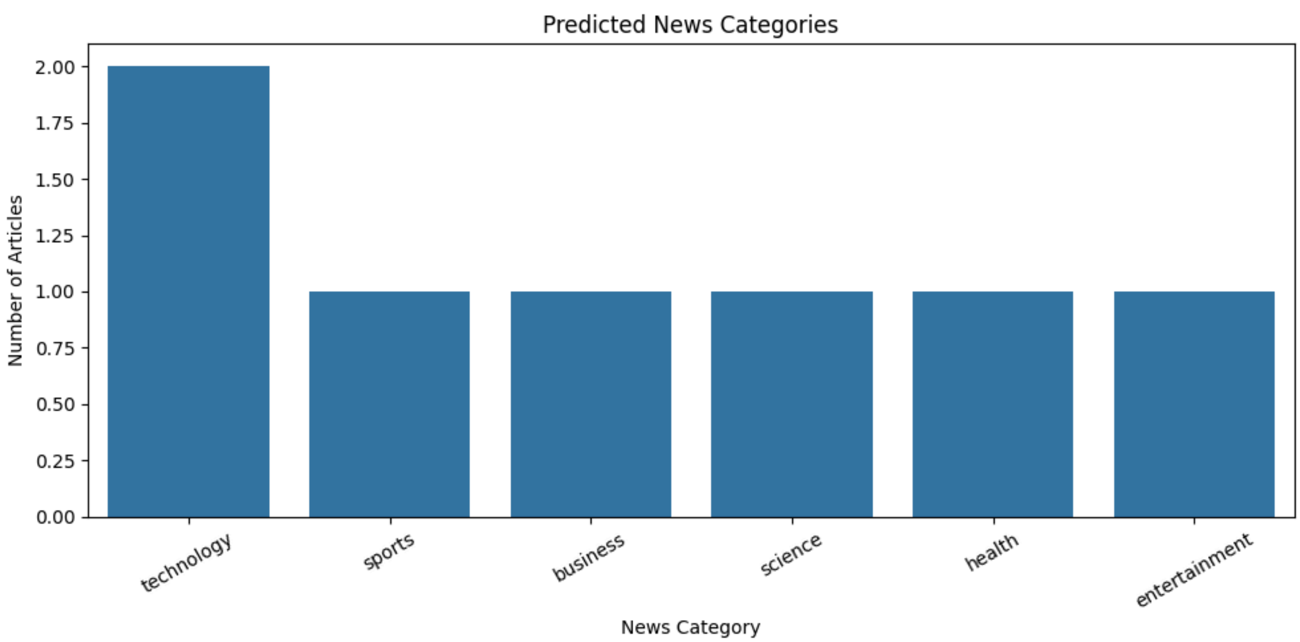
	text	category
0	The company launched a new artificial intellig...	technology
1	The football team won the championship after a...	sports
2	The company reported strong financial growth d...	business
3	Scientists discovered a new method for improvi...	science
4	The government announced a new policy for digi...	politics
5	Doctors developed a new approach to improve pa...	health
6	The movie received positive reviews from audie...	entertainment

```
results = []

for text in df["text"]:

    analysis = analyze_news(text)

    results.append({

        "News":
            text,

        "Actual Category":
            df.loc[
                len(results),
                "category"
            ],

        "Predicted Category":
            analysis["category"],

        "Confidence (%)":
            analysis["category_confidence"],

        "Sentiment":
            analysis["sentiment"],

        "Risk Level":
            analysis["risk_level"]
    })


results_df = pd.DataFrame(results)

display(results_df)
```

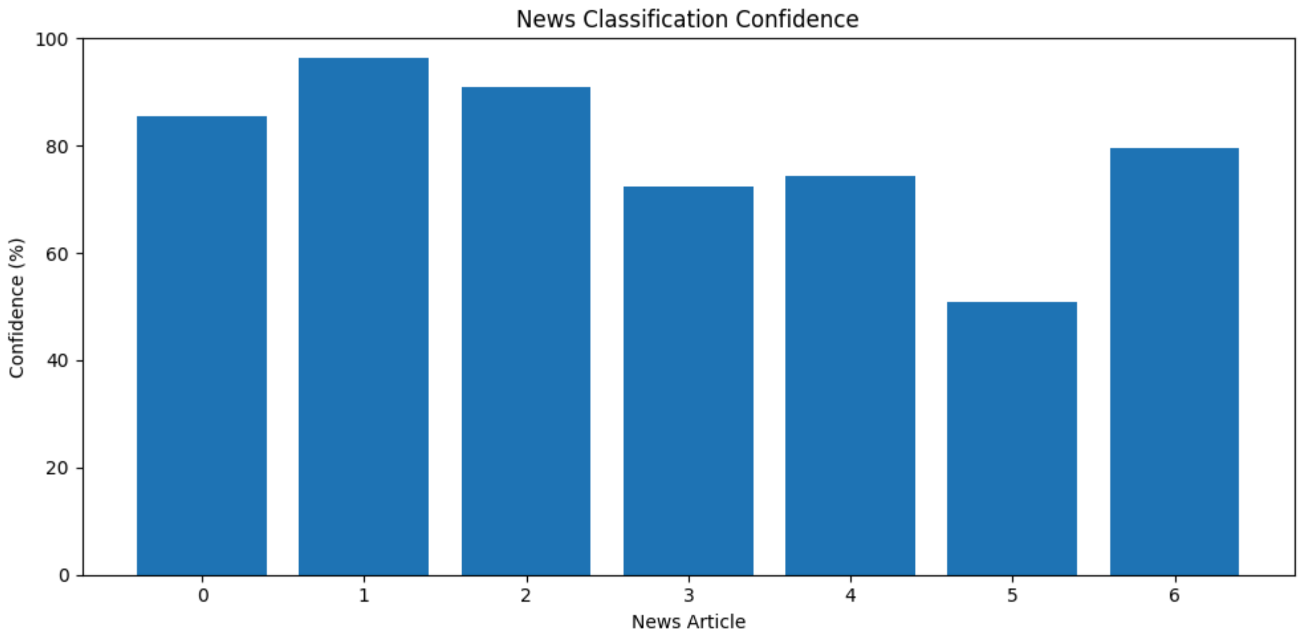
	News	Actual Category	Predicted Category	Confidence (%)	Sentiment	Risk Level
0	The company launched a new artificial intellig...	technology	technology	85.49	POSITIVE	LOW
1	The football team won the championship after a...	sports	sports	96.46	POSITIVE	LOW
2	The company reported strong financial growth d...	business	business	91.04	POSITIVE	LOW
3	Scientists discovered a new method for improvi...	science	science	72.42	POSITIVE	LOW
4	The government announced a new policy for digi...	politics	technology	74.30	POSITIVE	LOW
5	Doctors developed a new approach to improve pa...	health	health	50.78	POSITIVE	LOW
6	The movie received positive reviews from audie...	entertainment	entertainment	79.45	POSITIVE	LOW

```
plt.figure(figsize=(10, 5))

sns.countplot(
    data=results_df,
    x="Predicted Category"
)

plt.title(
    "Predicted News Categories"
)

```

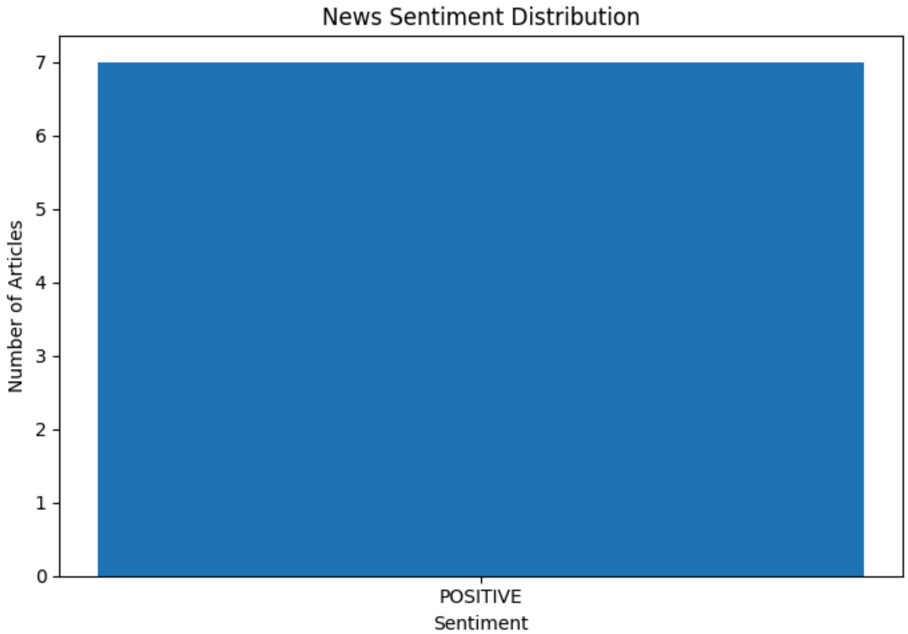
```python
plt.xlabel(
    "News Category"
)

plt.ylabel(
    "Number of Articles"
)

plt.xticks(rotation=30)

plt.tight_layout()

plt.show()
```



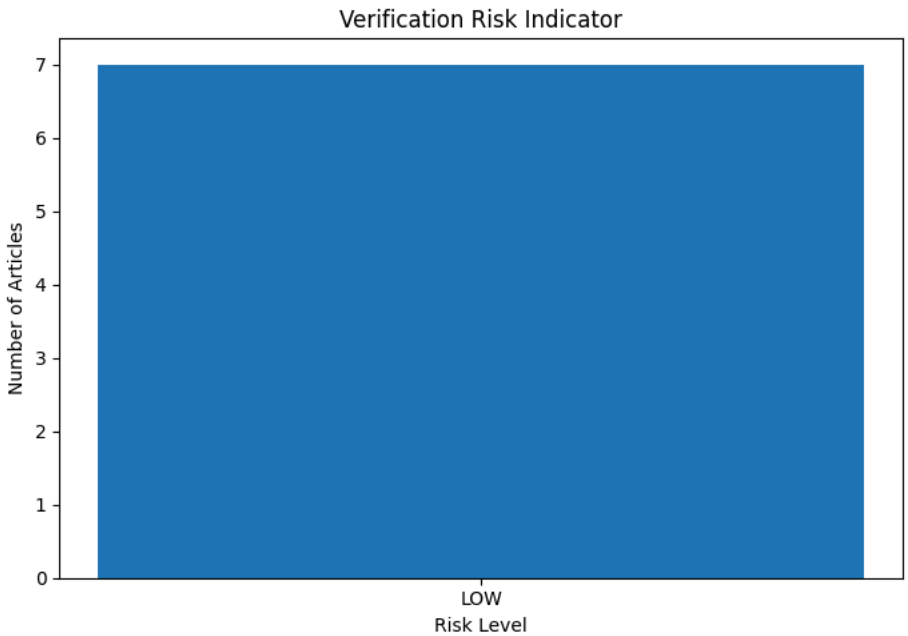

```python
plt.figure(figsize=(10, 5))

plt.bar(
    range(len(results_df)),
    results_df["Confidence (%)"]
)

plt.title(
    "News Classification Confidence"
)

plt.xlabel(
    "News Article"
)

plt.ylabel(
    "Confidence (%)"
)

plt.ylim(
    0,
    100
)

plt.tight_layout()

plt.show()
```




```
plt.figure(figsize=(7, 5))

sentiment_counts = results_df[
    "Sentiment"
].value_counts()

plt.bar(
    sentiment_counts.index,
    sentiment_counts.values
)

plt.title(
    "News Sentiment Distribution"
)

plt.xlabel(
    "Sentiment"
)

plt.ylabel(
    "Number of Articles"
)

plt.tight_layout()

plt.show()
```




```
plt.figure(figsize=(7, 5))

risk_counts = results_df[
    "Risk Level"
].value_counts()

plt.bar(
    risk_counts.index,
    risk_counts.values
)

plt.title(
    "Verification Risk Indicator"
)

plt.xlabel(
    "Risk Level"
)

plt.ylabel(
    "Number of Articles"
)

plt.tight_layout()

plt.show()
```




```
import gradio as gr


def gradio_analysis(text):

    result = analyze_news(text)

    if "error" in result:
        return result["error"]

    signals = (
        ", ".join(result["risk_signals"])
        if result["risk_signals"]
        else "No major linguistic risk signals detected."
    )

    keywords = (
        ", ".join(result["keywords"])
        if result["keywords"]
        else "No keywords detected."
    )

    output = f"""
#  AI News Intelligence Report

---

##  News Category

### {result["category"].upper()}

**Confidence:** {result["category_confidence"]}%

---

##  Sentiment

### {result["sentiment"]}

**Confidence:** {result["sentiment_confidence"]}%

---

##  Important Keywords

{keywords}

---

```

```
##  Verification Risk Indicator

### {result["risk_level"]}

**Risk Score:** {result["risk_score"]}%

**Detected Linguistic Signals:**

{signals}

---

##  AI Generated Summary

{result["summary"]}

---

###  Disclaimer

The risk indicator identifies potentially sensational
linguistic patterns and is **not a factual determination**.
Important information should always be verified using
reliable and independent sources.
"""

    return output


demo = gr.Interface(

    fn=gradio_analysis,

    inputs=gr.Textbox(
        lines=12,
        placeholder=
        "Paste a news article or headline here..."
    ),

    outputs=gr.Markdown(),

    title=
    " AI News Intelligence & Risk Analyzer",

    description=
    """
    Transformer-based AI system for news classification
```